

VLAN and Inter-VLAN Routing Using FortiGate

Project Team

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1. Introduction

This project demonstrates the configuration of Virtual Local Area Networks (VLANs) and inter-VLAN routing using a Layer 2 switch and a FortiGate firewall. The primary objectives are to segment a network for enhanced security, enable controlled communication between VLANs, and enforce firewall policies to regulate traffic. The setup is tested to ensure proper functionality and security.

2. Project Overview

The project focuses on:

- Configuring VLANs on a Layer 2 switch to segment the network.
- Using a FortiGate firewall to perform inter-VLAN routing.
- Applying firewall policies to control traffic flow between VLANs.
- Verifying communication through testing and monitoring.

Objectives:

- Achieve network segmentation for security and efficiency.
- Enable secure and controlled inter-VLAN communication.
- Validate the configuration using ping tests, routing tables, and logs.

3. Network Topology

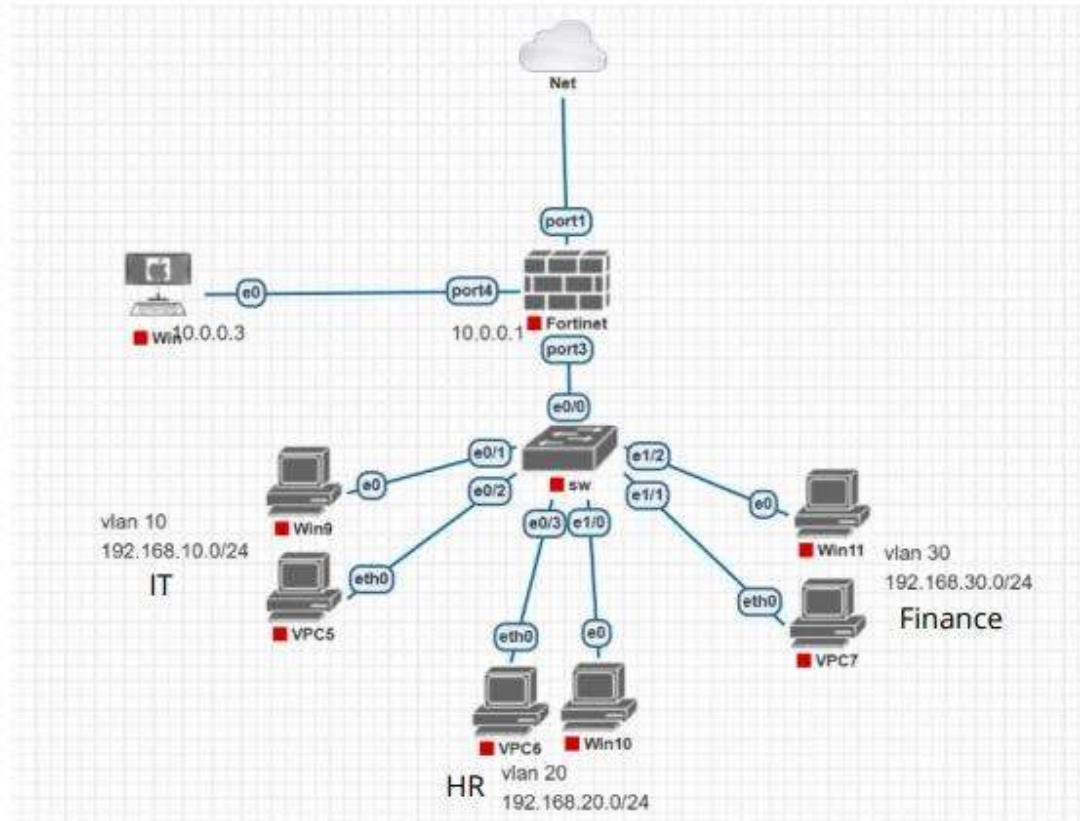
The network consists of:

- **Three VLANs:** VLAN 10 (HR), VLAN 20 (Finance), VLAN 30 (Guest).
- **Layer 2 Switch:** Connects end devices to VLANs and links to FortiGate via a trunk port.
- **FortiGate Firewall:** Handles inter-VLAN routing and enforces firewall policies.

IP Addressing (example):

- VLAN 10 (HR): 192.168.10.0/24
- VLAN 20 (Finance): 192.168.20.0/24
- VLAN 30 (Guest): 192.168.30.0/24

NETWORK TOPOLOGY



4. VLANs Overview

4.1 What is a VLAN?

A VLAN (Virtual Local Area Network) logically segments a physical network, grouping devices based on function or security needs without requiring separate hardware.

4.2 Why Use VLANs?

- **Security:** Isolates sensitive data (e.g., HR from Guest).
- **Performance:** Reduces broadcast traffic.
- **Management:** Simplifies network organization by department or role.

4.3 Benefits of VLANs

- Enhanced security through isolation.
- Improved network efficiency.
- Flexible device grouping.

4.4 What is Inter-VLAN Routing?

Inter-VLAN routing enables communication between VLANs via a Layer 3 device (FortiGate in this project). It allows controlled data exchange while maintaining segmentation.

5. VLAN Configuration on Switch

The switch is configured to support three VLANs and a trunk port to the FortiGate.

5.1 Create VLANs

- VLAN 10: HR
- VLAN 20: Finance
- VLAN 30: Guest

Example CLI (Cisco Switch):

```
vlan 10 name HR vlan 20 name Finance vlan 30 name Guest
```

5.2 Configure Access Ports

Assign switch ports to VLANs for end devices.

Example:

```
interface FastEthernet0/1 switchport mode access switchport access  
vlan 10 interface FastEthernet0/2 switchport mode access switchport  
access vlan 20 interface FastEthernet0/3 switchport mode access  
switchport access vlan 30
```

5.3 Configure Trunk Port

Set up a trunk port to carry VLAN traffic to FortiGate.

Example:

```
interface GigabitEthernet0/1 switchport mode trunk switchport trunk
allowed vlan 10,20,30
```

Note: Replace with your switch's specific commands/model (e.g., Cisco Catalyst, HP ProCurve).

6. FortiGate Configuration

The FortiGate firewall is configured for inter-VLAN routing and IP address assignment.

6.1 Create VLAN Interfaces

- Access FortiGate GUI: <http://10.0.0.1> (or your FortiGate IP).
- Navigate to **Network > Interfaces > Create New > VLAN**.
- Configure:
 - VLAN 10: ID 10, IP 192.168.10.1/24, Parent Interface: Port 3
 - VLAN 20: ID 20, IP 192.168.20.1/24, Parent Interface: Port 3
 - VLAN 30: ID 30, IP 192.168.30.1/24, Parent Interface: Port 3

Example CLI:

```
config system interface edit "vlan10" set vdom "root" set ip
192.168.10.1 255.255.255.0 set vlanid 10 set interface "port3" next
edit "vlan20" set vdom "root" set ip 192.168.20.1 255.255.255.0 set
vlanid 20 set interface "port3" next edit "vlan30" set vdom "root" set
ip 192.168.30.1 255.255.255.0 set vlanid 30 set interface "port3" next
end
```

6.2 Enable DHCP (Optional)

Configure DHCP servers for client devices.

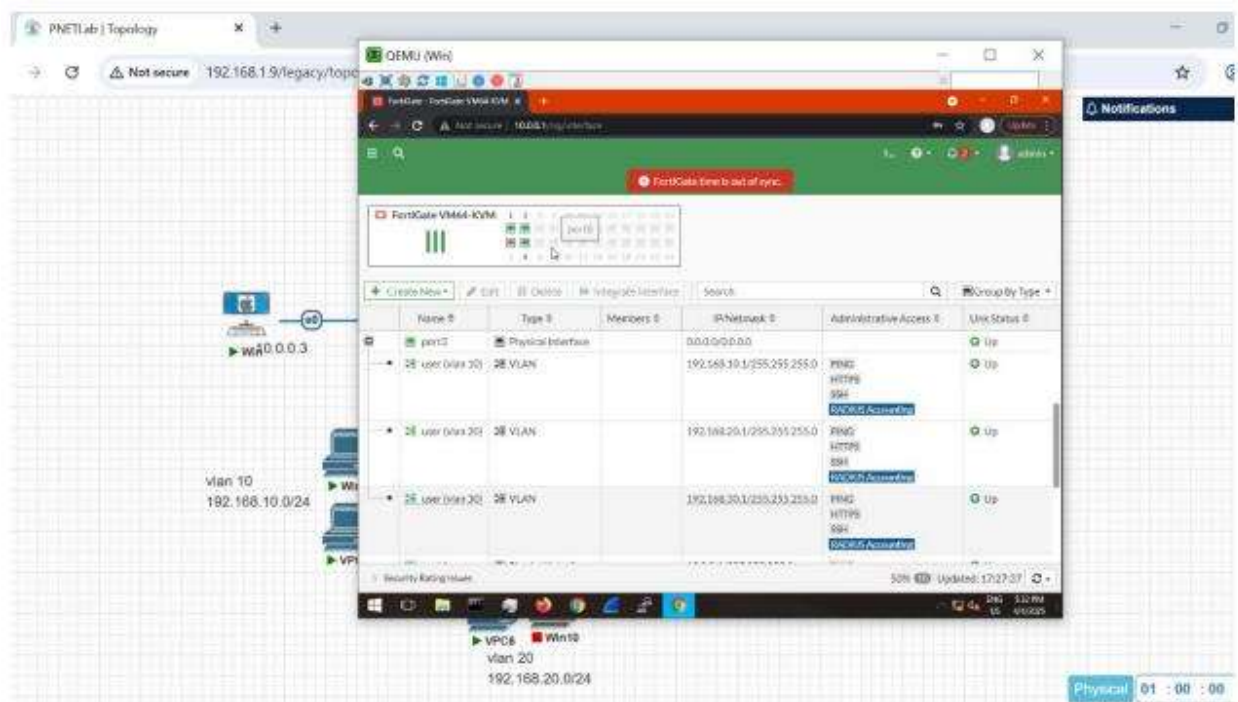
- VLAN 10: 192.168.10.100–192.168.10.200

- VLAN 20: 192.168.20.100–192.168.20.200
- VLAN 30: 192.168.30.100–192.168.30.200

GUI: Network > Interfaces > [VLAN] > Enable DHCP Server.

6.3 Verify Subinterfaces

Subinterfaces for VLANs 10, 20, and 30 are created on FortiGate's port 3 (trunk).



7. FortiGate Firewall Policy

Firewall policies control traffic between VLANs and to external networks.

7.1 What is a Firewall Policy?

A firewall policy defines rules for traffic based on:

- Source (e.g., VLAN 10)
- Destination (e.g., VLAN 20)
- Service (e.g., HTTP, DNS)

- Action (ACCEPT or DENY)

7.2 Purpose in This Project

- Allow specific inter-VLAN communication (e.g., HR to Finance).
- Block unauthorized access (e.g., Guest to HR).
- Permit internet access for select VLANs.
- Restrict traffic to approved protocols.

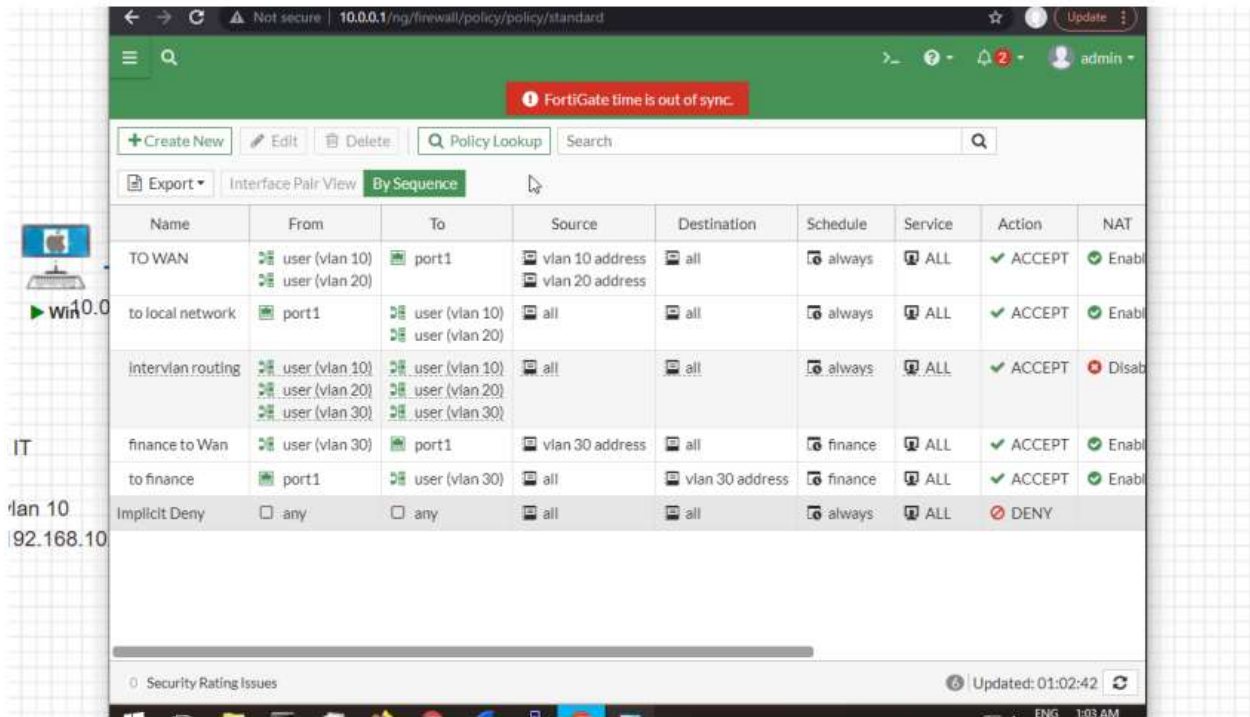
7.3 Example Policies

Policy ID	Name	Incoming Interface	Outgoing Interface	Source	Destination	Service	Action	Notes
1	HR_to_Finance	VLAN 10	VLAN 20	All	All	HTTP, DNS	ACCEPT	Yes
2	Block_Guest_to_HR	VLAN 30	VLAN 10	All	All	All	DENY	No
3	Finance_to_Internet	VLAN 20	WAN	All	All	HTTP, HTTPS, DNS	ACCEPT	Yes

GUI Configuration:

- Navigate to **Policy & Objects > IPv4 Policy > Create New**.
- Set parameters as above.

FORTIGATE POLICY



8. Verification and Testing

The setup was validated through multiple tests.

8.1 Ping Test

- **Test Case:** Ping from VLAN 10 (e.g., 192.168.10.100) to VLAN 20 (e.g., 192.168.20.100).
- **Command:**

```
ping 192.168.20.100
```

- **Expected Result:** Success if policy allows; failure if blocked (e.g., Guest to HR).

8.2 Routing Table Check

- Verify FortiGate routing configuration.
- **Command:**

```
get router info routing-table all
```

- **Expected Output:** Routes for 192.168.10.0/24, 192.168.20.0/24, 192.168.30.0/24.

8.3 Traffic Logs

- Monitor traffic to confirm policy enforcement.
- **GUI:** Log & Report > Local Traffic Log.
- **Expected:** Logs show allowed traffic (e.g., HTTP from VLAN 10 to 20) and denied traffic (e.g., VLAN 30 to 10).

Placeholder: Insert sample outputs (e.g., ping results, routing table, logs).

9. Conclusion

- VLANs were successfully configured on the switch, segmenting the network into HR, Finance, and Guest VLANs.
- Inter-VLAN routing was implemented using FortiGate, enabling controlled communication.
- Firewall policies enhanced security by allowing specific traffic and blocking unauthorized access.
- Tests confirmed proper functionality and policy enforcement.

10. Future Improvements

- Implement more granular policies (e.g., application-specific rules).
- Add VLANs for additional departments.
- Explore FortiGate features like Intrusion Prevention System (IPS) or SSL inspection.